

Year 8 Science

Sedimentary Rocks - Weathering

Sediments are particles of large rocks – including mountains, hills and cliffs – that have been **weathered** and eroded over time. There are several ways in which weathering occurs: chemical weathering, physical weathering (onion-skin/exfoliation and freeze-thaw) and biological weathering.

Your task:

1. Use the websites listed to find out: **what** each type is, and **where** they are likely to occur.

http://www.geolsoc.org.uk/ks3/gsl/education/resources/rockcycle/page3461.html

- Use the underlined hyperlinks to further explore the website.
- Watch the freeze-thaw animation for a better example of this slow process

http://www.bbc.co.uk/bitesize/ks3/science/environment_earth_universe/rock_cycle/revision/6/

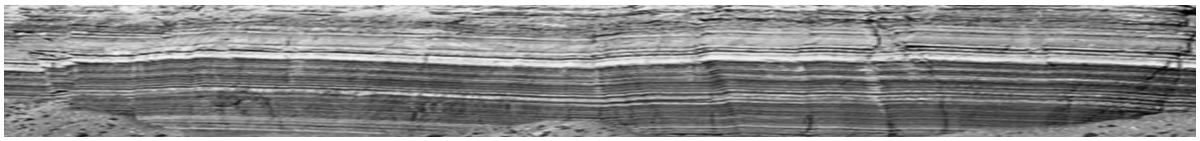
Chemical Weathering

1. In a few sentences, describe how chemical weathering of rocks occurs.
   Rain is slightly acidic, which reacts with minerals in rock and weathers it
2. In which climates is chemical weathering most likely to occur?

   Hot/dry Cold/icy **Humid/rainy**

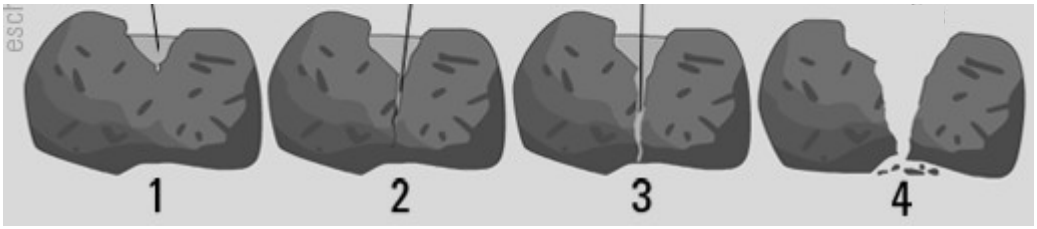
   Give reason for your answer. *Acid is often in rain water*
3. The picture below shows the effects of chemical weathering. Label or caption this picture to show how this statue has been affected by chemical weathering.





Physical weathering – Freeze-thaw

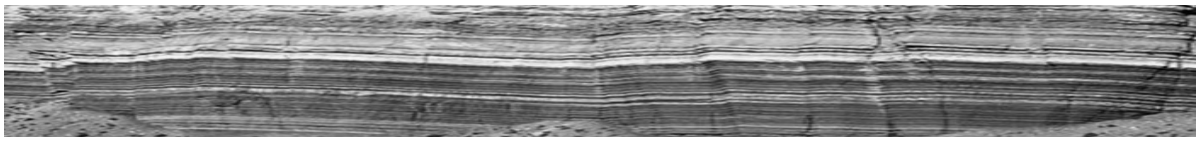
1. In a few sentences, list the steps of freeze-thaw weathering. Use the diagram below as a guide
   a. Water fills small cracks in rock.
   b. When water freezes, it expands, pushing against the rock. The crack gets bigger
   c. The ice thaws. More water can now fill the bigger crack.
   d. The cycle repeats until the crack is so big that rock particles break off



2. In which climates is freeze-thaw weathering most likely to occur?

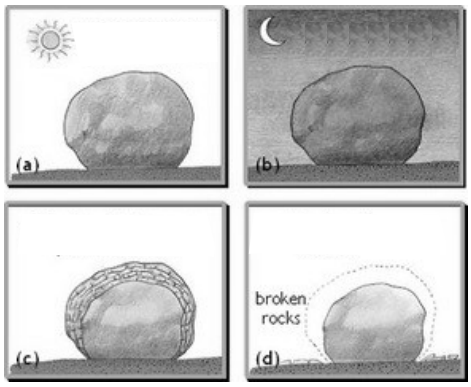
   Hot/dry **Cold/icy** Humid/rainy

   Give reason for your answer. *Cold climates mean water will more likely freeze*



Physical weathering – exfoliation

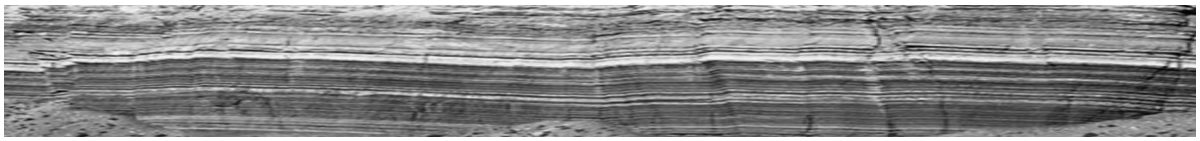
1. In a few sentences, list the steps of exfoliation. Use the diagram below as a guide.
   a. *Rocks expands when they get hot*
   b. *They contract when they get cold*
   c. *This repeats over time and cracks eventually form*
   d. *The cracks get larger to the point where rock breaks off (weathered)*



2. In which climates is exfoliation most likely to occur?

   Hot/dry Cold/icy Humid/rainy

   Give reason for your answer. *Heat is needed to expand the rock*



Biological weathering

1. In a few sentences, describe biological weathering

   Trees can grow in rocky terrain. Their roots can break rock, resulting in rock sediments

2. In which climates is biological weathering most likely to occur?

   Hot/dry Cold/icy Humid/rainy

   Give reason for your answer.

   _____

   _____

3. Draw and label a picture, or a set of pictures, below to show biological weathering. Remember, your pictures should show how large rocks are broken into small particles.